



FedE-ator

FedE-ator: How Much Energy Does Federated Learning “Eat” in Distributed Systems?

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Motivation

- Mostly ML studies optimize **accuracy** & ignores **energy cost**^[1]
 - FL energy spans **multiple clients** + **communication rounds** → inherently harder to measure than centralized ML
 - No unified framework: ad-hoc SW (RAPL, NVML) vs. HW meters → **non-reproducible**, non-comparable
- ⇒ Need a framework to learn the tradeoff between energy and FL performance in distributed environments.

Contributions

- FedE-ator**: reproducible energy measurement tool for FL - synchronized HW+SW monitoring, configurable DVFS, supports multiple FL frameworks
- Analysis framework**: studies energy-performance tradeoff across FL frameworks, ML model architectures, and CPU/GPU frequency settings

FedE-ator Framework

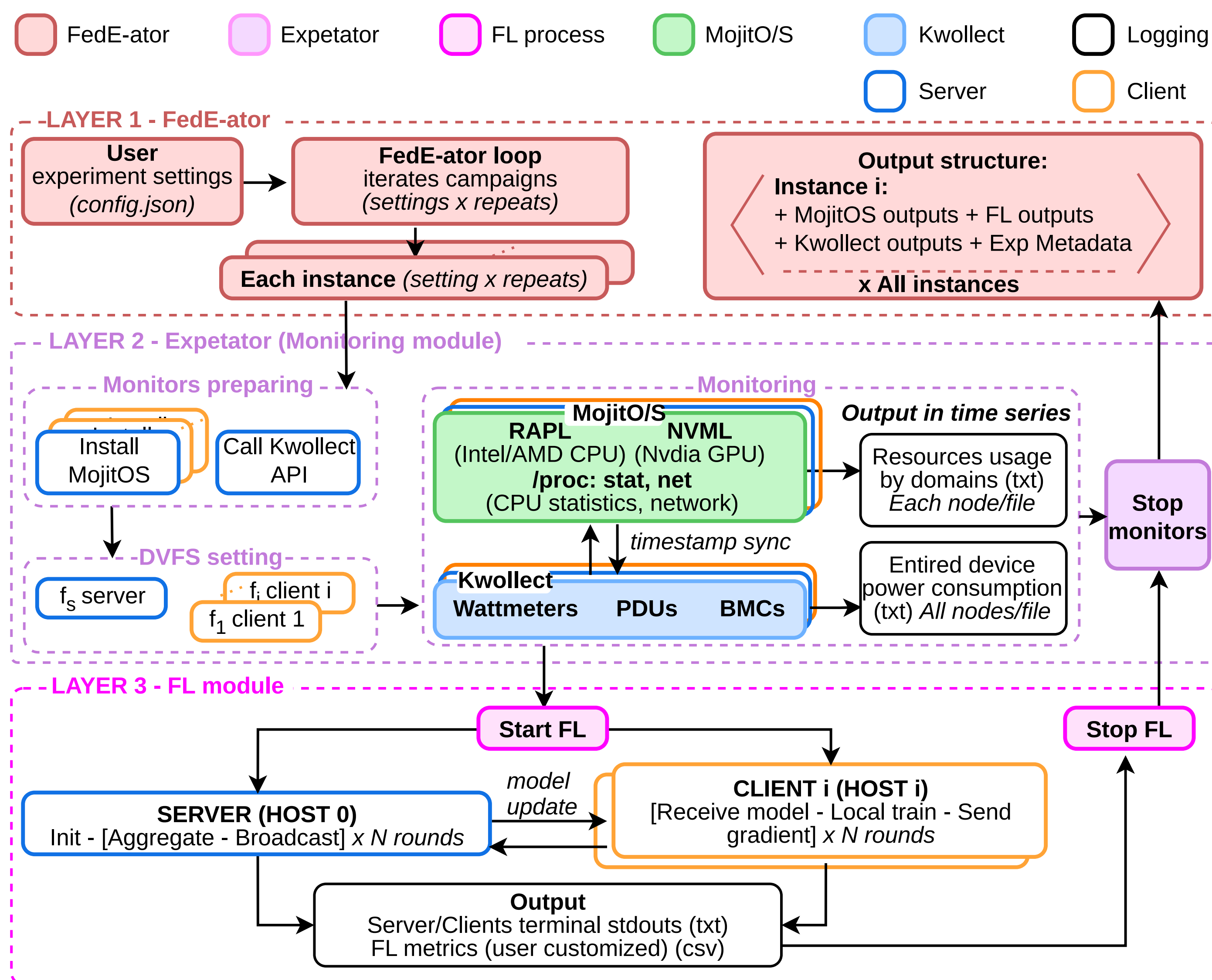


Figure 1. FedE-ator workflow: Expetator, MojitoS, and Kwolect as 3rd-party tools.

Three-layer architecture:

- FedE-ator Core**: reads JSON config, orchestrates campaigns (settings × repetitions), stores output
- Expetator (Monitoring)**: applies both SW (MojitoS^[2]) + HW (Kwolect) and Synchronized via shared timestamps.
- FL Module**: SSH-based execution on server/client nodes

Experiments design

Platform:

- Grid'5000/SLICES-FR^[3]
- Selected clusters: nova (CPU), gemini (GPU), gros (CPU)

Scenarios:

- SC1 - Heterogeneous**: CPU+GPU nodes, 3 freqs, 2 repetitions.
- SC2 - Cross-framework**: CPU nodes, 2 FL frameworks, 3 models, 4 freqs, 5 repetitions

Benchmark: 2 FL frameworks (Flower & FedScale), FedAvg strategy, evaluation in client-side by avg. accuracy, use CIFAR-10 dataset (IID split).

Experiments result

SC1 - Heterogeneous

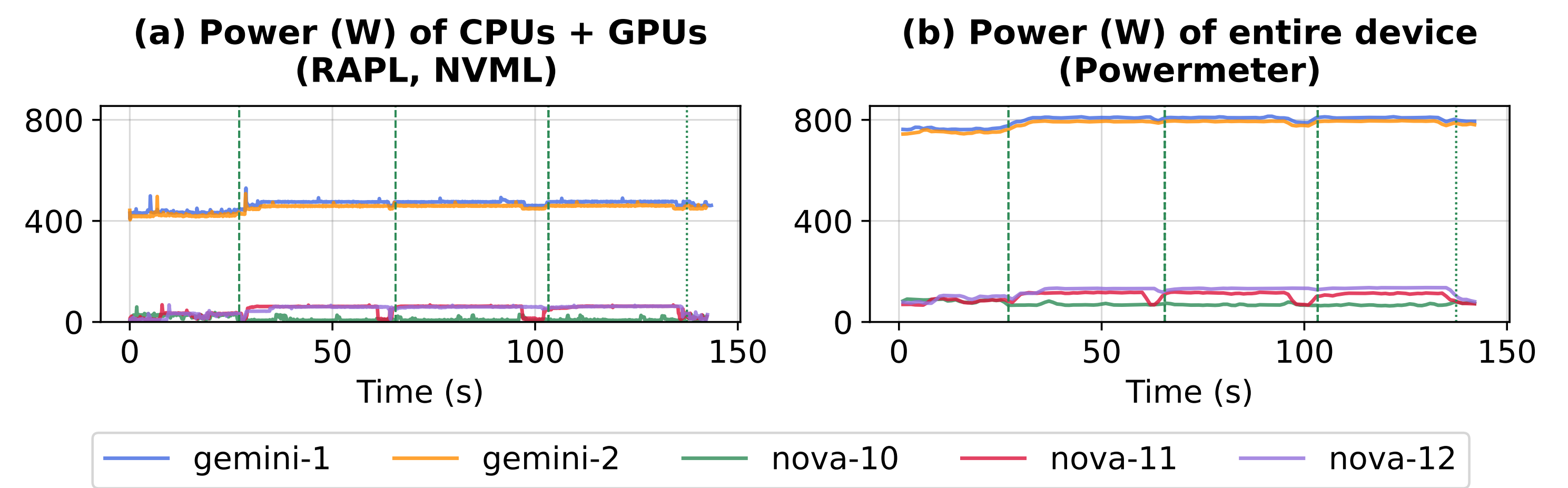


Figure 2. SW vs. HW in SC1 at $f_{\max}$. Power meter reports $\approx 2\times$ higher values, capturing full-system consumption (ie., motherboard, NIC, fan).

Table 1. Frequency Impact on Energy & Accuracy

Freq. [GPU, CPU] (GHz)	[0.135, 1.2]	[0.832, 1.7]	[1.53, 2.1]
Time monitored / FL training (s)	250 / 231	158 / 145	134 / 112
Mean power (W)	350	359	377
Total energy CPUs+GPUs (kJ)	231	152	140
Total energy entire system (kJ)	442	286	254
Accuracy (3 rounds)	0.556	0.557	0.563

SC2 - Energy and Performance Tradeoff Cross-framework

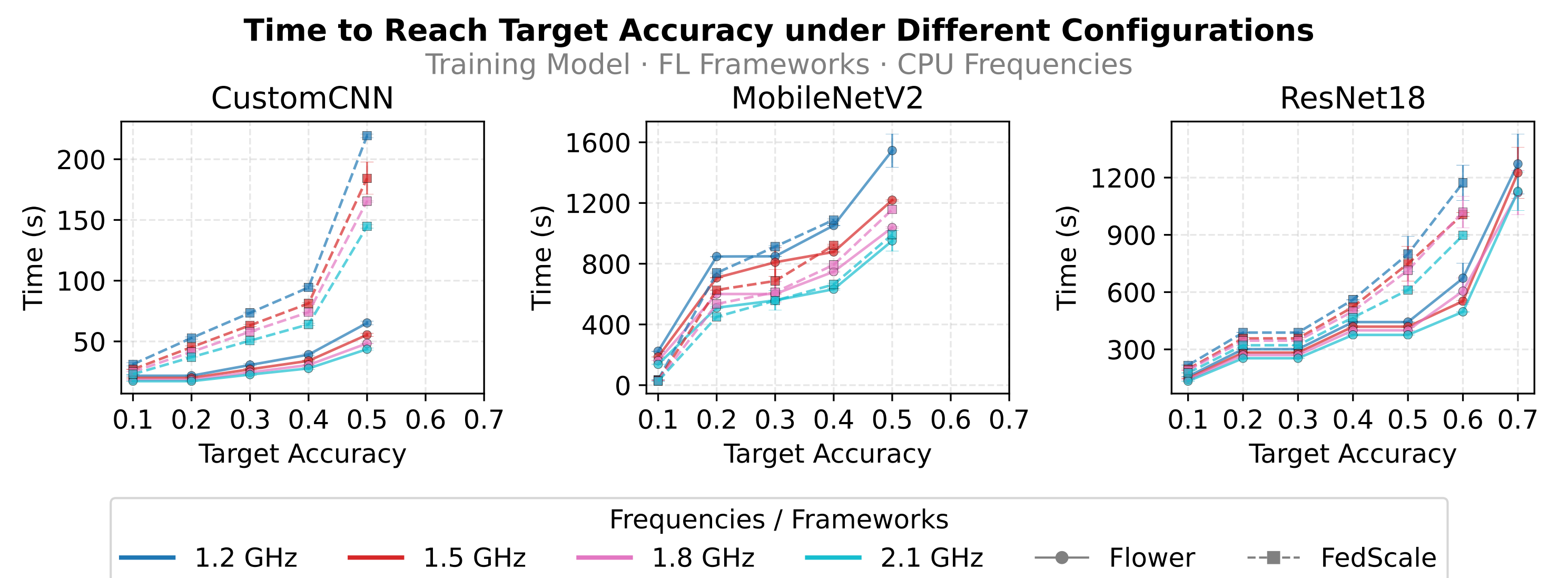


Figure 3. Time to reach target accuracy across CPU frequencies, models, and FL frameworks (SC2). Mean over 5 runs, error bars = std.

Energy Consumption to Reach Target Accuracy under Different Configurations

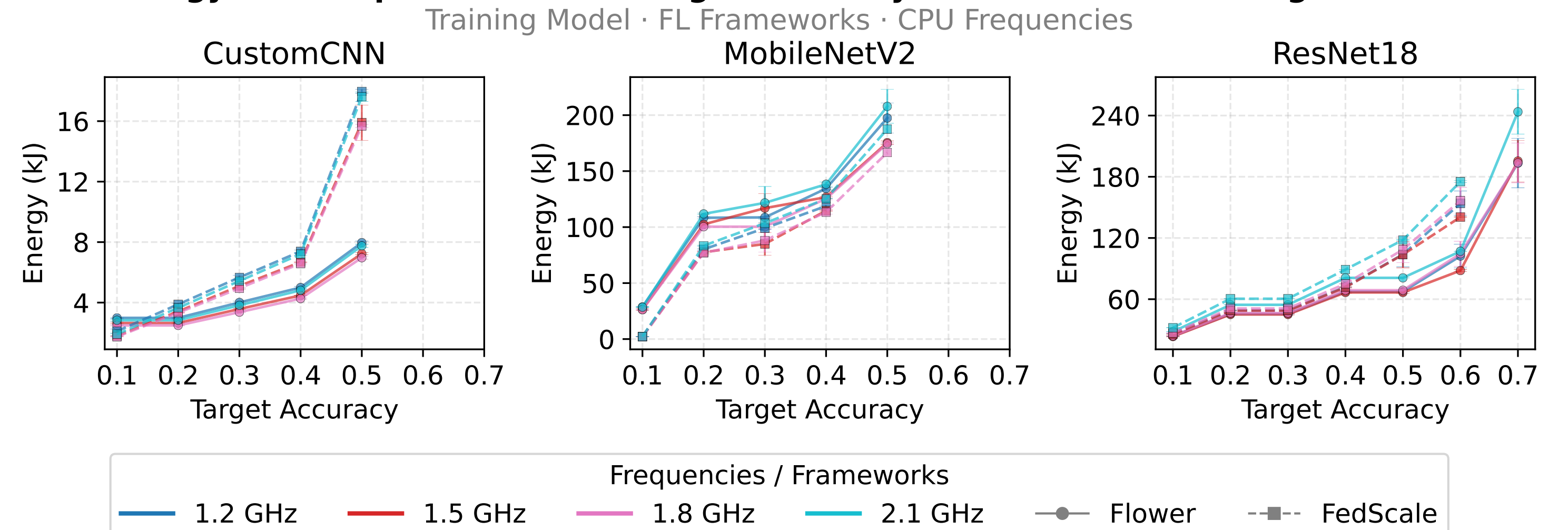


Figure 4. Energy to reach target accuracy across CPU frequencies, models, and FL frameworks (SC2). Mean over 5 runs, error bars = std.

Conclusion and Future works

Conclusions:

- Proposed FedE-ator framework - measuring the energy consumption of FL in distributed systems. Validated on Grid'5000.
- Proposed a framework to analyze the energy-performance tradeoff.

Future work:

- Deploy FedE-ator beyond Grid'5000 to other infrastructures.
- Extend the framework with DVFS online policy.

KEY REFERENCES

- [1] Schwartz, Roy, et al. “Green ai.” *Communications of the ACM* 63.12 (2020).
- [2] [Software] Georges Da Costa: MojitoS (Feb 2026), <https://hal.science/hal-03453537>
- [3] [Website] <https://www.grid5000.fr/>

ACKNOWLEDGEMENT AND PARTNERS



MORE INFORMATION



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